



Querying Data With Starburst SQL

Lesson objectives

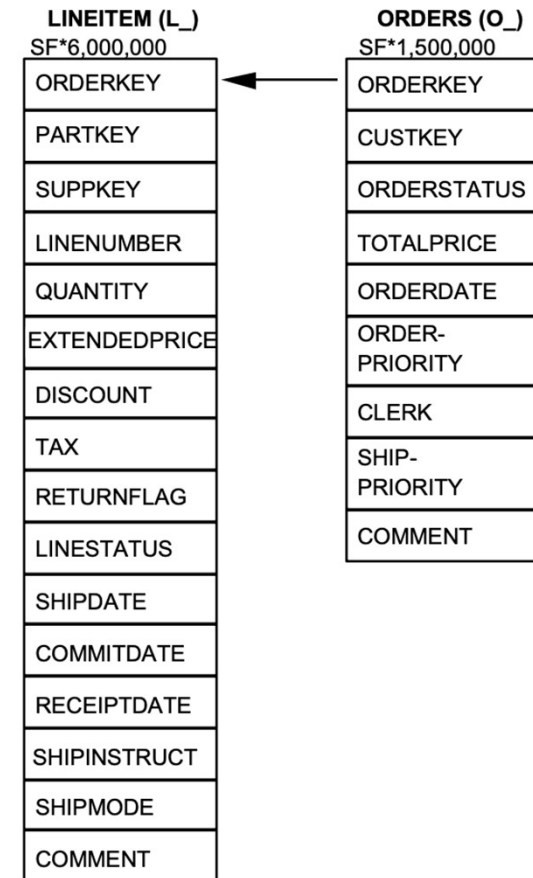
Querying Data With Starburst SQL

1. Review SQL basics in Starburst.
1. Write and run SQL queries in the query editor.
1. Filter, aggregate, and join data from multiple sources.

Expected SQL knowledge

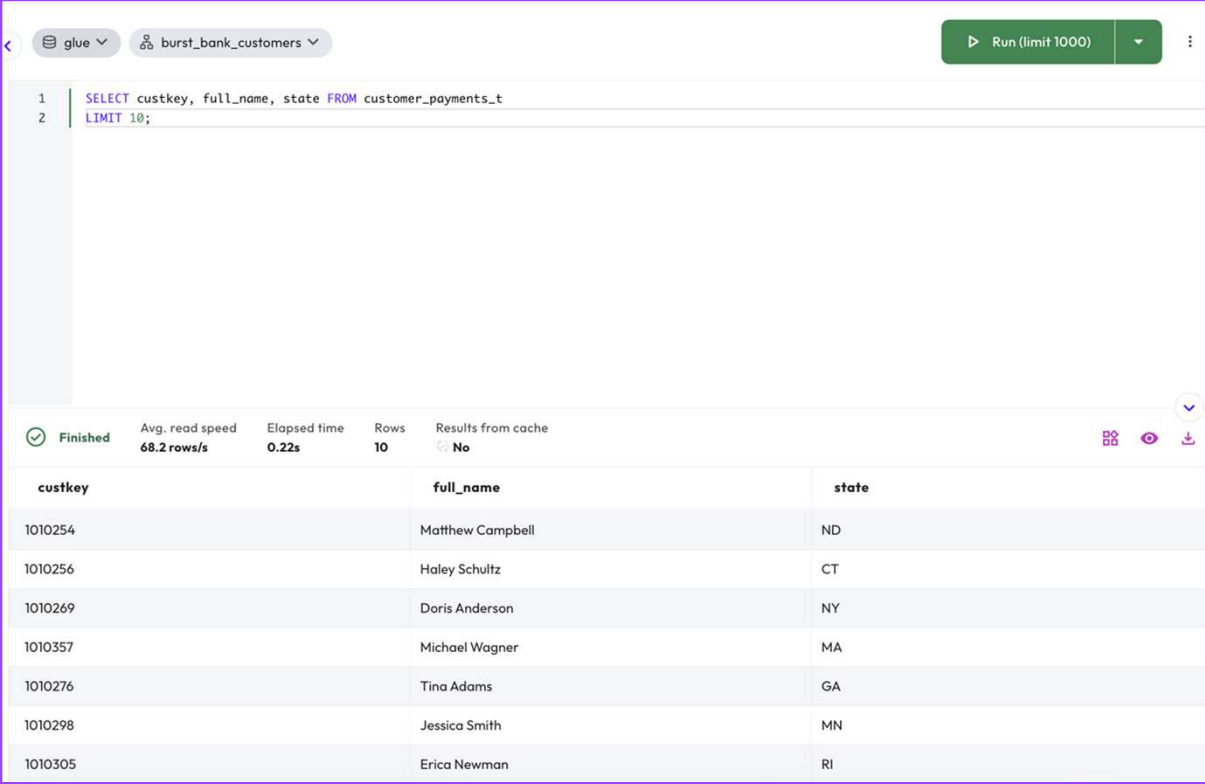
Projecting, filtering, ordering, aggregating, and joining

- Basic `SELECT` statement on specified columns in a table
- Using `WHERE` clauses to limit the rows of data
- Sorting the results with `ORDER BY`
- Performing aggregations such as `COUNT`, `MAX`, `SUM` and `AVG`
 - Across the entire table population
 - For each rolled up section identified in a `GROUP BY` clause
- Combining two (or more) tables with the `JOIN` keyword



Starburst SQL

- ANSI-SQL compliant
- Multiple data sources; one language
- Query engine translates for each data source
- Reference guide [here](#)



The screenshot displays the Starburst SQL interface. At the top, there are tabs for 'glue' and 'burst_bank_customers'. A green button labeled 'Run (limit 1000)' is in the top right. The query editor shows two lines of SQL: '1 SELECT custkey, full_name, state FROM customer_payments_t' and '2 LIMIT 10;'. Below the editor, a status bar indicates 'Finished' with a green checkmark, 'Avg. read speed 68.2 rows/s', 'Elapsed time 0.22s', 'Rows 10', and 'Results from cache No'. The results are shown in a table with three columns: 'custkey', 'full_name', and 'state'. The table contains 10 rows of data.

custkey	full_name	state
1010254	Matthew Campbell	ND
1010256	Haley Schultz	CT
1010269	Doris Anderson	NY
1010357	Michael Wagner	MA
1010276	Tina Adams	GA
1010298	Jessica Smith	MN
1010305	Erica Newman	RI

What can we do with multiple catalogs?

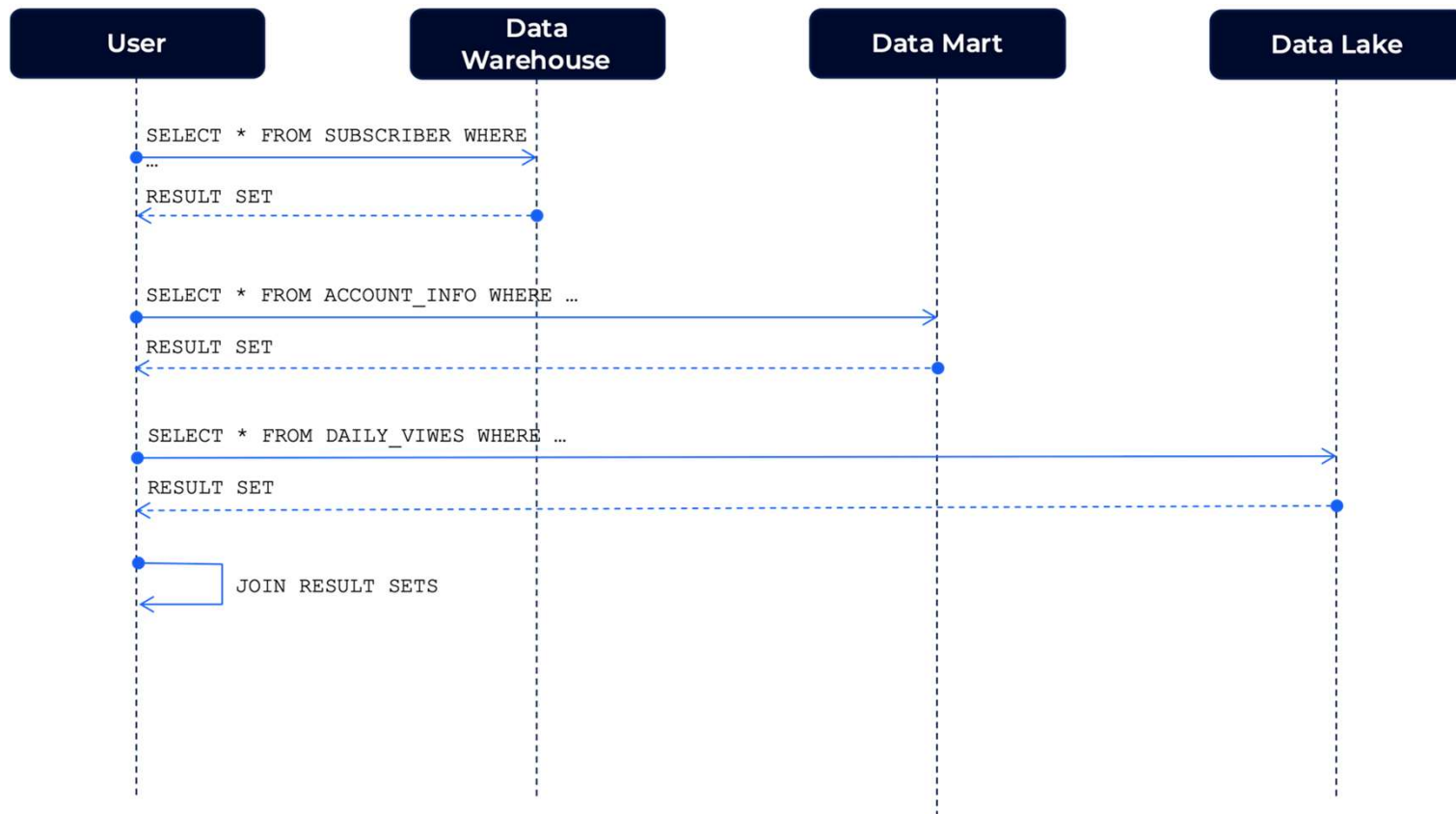
Single point of access for a variety of data sources

- Same UI/CLI/API for access all of your data.
- Configure security and governance in one place.

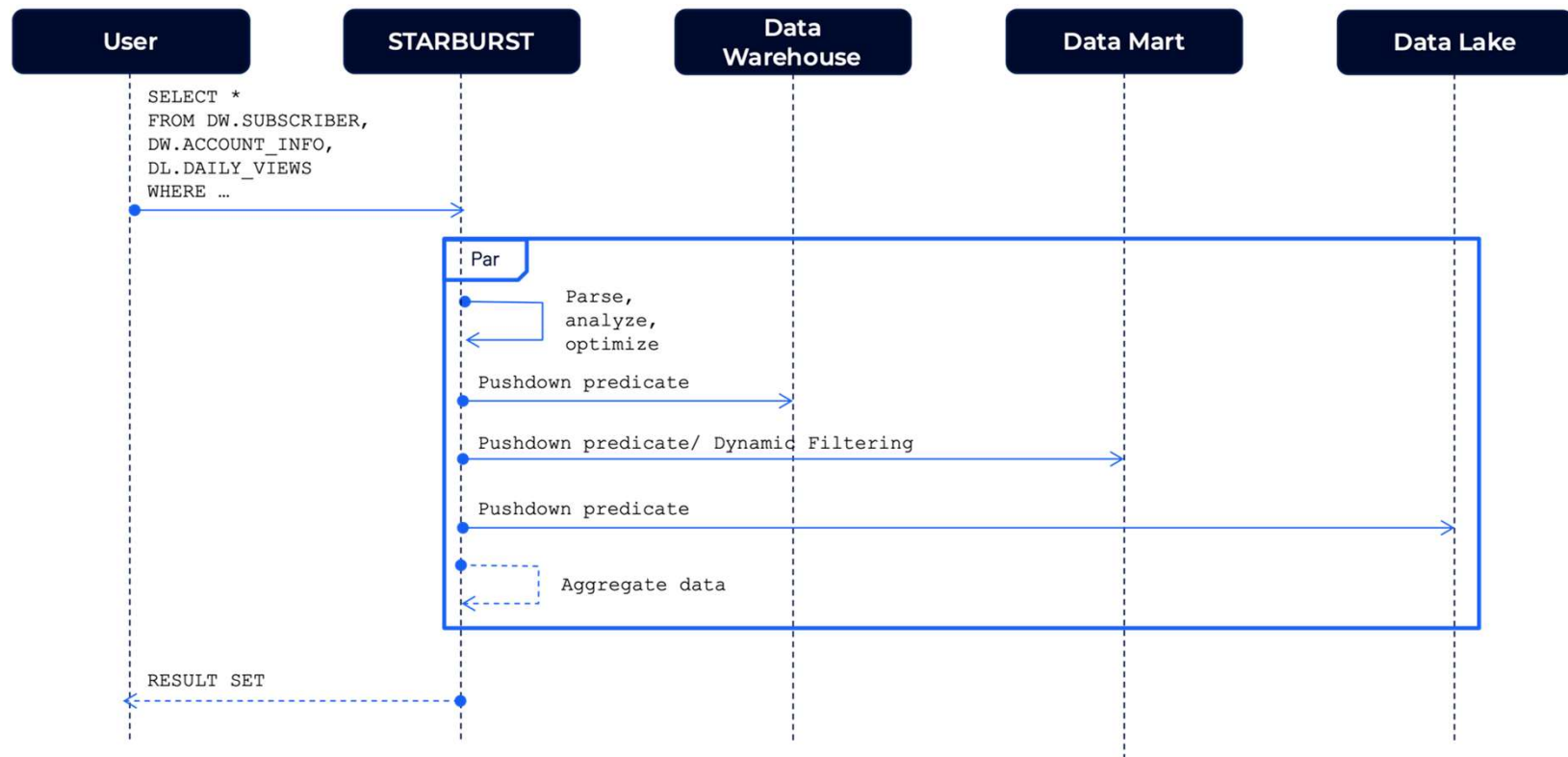
Query federation

- Access data from multiple systems within a single query
 - For example, join historic log data stored in an S3 object storage with customer data stored in a MySQL database.

Query federation the traditional way



Query federation the Starburst way



Instructor demonstration

Joining sales data from a cloud
warehouse with marketing
data in object storage (15 mins)

Hands-On Exercise

Write queries to answer key
business questions (20
minutes)

Lesson summary

Querying Data With Starburst SQL

1. Starburst is ANSI-SQL compliant.
2. You can query all of your data sources using the same SQL.
3. Query federation saves time that would have been spent searching for and moving data.



Data Products for Analysts